

Hermitian Operators

$$\langle \psi | \hat{M} | \phi \rangle = \langle \underbrace{\psi | \hat{M} \phi} \rangle$$

Inner Product
or
Dot Product

$$\underline{|\psi| |\hat{M}\phi| \cos \theta}$$

Now do we have some $\hat{N}$ which gives:

$$\langle \underbrace{\hat{N} \psi | \phi} \rangle = \langle \psi | \hat{M} \phi \rangle$$

Hermitian Adjoint

$$\boxed{N = M^\dagger}$$

Some Properties : ① $(A^\dagger)^\dagger = A$

$$\text{② } (A+B)^\dagger = A^\dagger + B^\dagger$$

Important $\rightarrow$ ③ $(AB)^\dagger = B^\dagger A^\dagger$

$$\text{④ } c^\dagger = c^* : \text{Scalar.}$$

$\Rightarrow$ What is † of a ket $|\psi\rangle$?

$$\langle\psi|\phi\rangle^\dagger = \langle\psi|\phi\rangle^* : \text{Inner product is a } \underline{\text{Scalar.}}$$
$$= \langle\phi|\psi\rangle$$

OR

$$\langle\psi|\phi\rangle^\dagger = |\phi\rangle^\dagger \langle\psi|^\dagger$$

$$|\phi\rangle^\dagger \langle\psi|^\dagger = \langle\phi|\psi\rangle : \text{should be true!}$$

So :

$$|\phi\rangle^\dagger = \langle\phi|$$

and

$$\langle\psi|^\dagger = |\psi\rangle$$

$\text{bra}^\dagger = \text{ket}$
 $\text{ket}^\dagger = \text{bra}$

What is (Observable) † : $E^\dagger$?

$$|E_i\rangle = \text{definite states} \quad \& \quad E_i = \text{measured value}$$

$\downarrow$
Eigenvector

$\downarrow$ value
Eigenvalue

- $E_i \in \mathbb{R}$
- $\text{span}(|E_i\rangle) = V$
- $\langle E_i | E_j \rangle = \delta_{ij}$: Orthonormal

$\Rightarrow$ Say : $\hat{E}|\psi\rangle \Rightarrow \hat{E}(\sum c_i |E_i\rangle)$

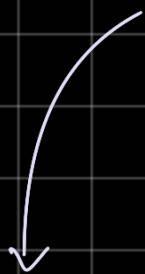
$$= \sum c_i \hat{E} |E_i\rangle \quad \& \quad c_i = \langle E_i | \psi \rangle$$

$$= \sum_i \langle E_i | \psi \rangle \hat{E} | E_i \rangle$$

$$= \sum_i | E_i \rangle \langle E_i | \psi \rangle$$

$$\hat{E} \psi = \left(\sum_i | E_i \rangle \langle E_i | \right) \psi$$

$$\hat{E} = \sum_i | E_i \rangle \langle E_i |$$



or

$$\hat{x} = \int dx \, x | x \rangle \langle x |$$

$$\hat{E}^\dagger = \left(\sum_i | E_i \rangle \langle E_i | \right)^\dagger$$

$$= \sum_i \left(| E_i \rangle \langle E_i | \right)^\dagger$$

$$= \sum_i \langle E_i |^\dagger | E_i \rangle^\dagger E_i^\dagger$$

$$= \sum_i | E_i \rangle \langle E_i | E_i^*$$

$$= \sum_i | E_i \rangle \langle E_i | = \hat{E}$$

$$\hat{E}^\dagger = \hat{E}$$

Hermian Operators

All observables are Hermian operators!

$$\langle \hat{E} \psi | \phi \rangle = \langle \psi | \hat{E} \phi \rangle$$